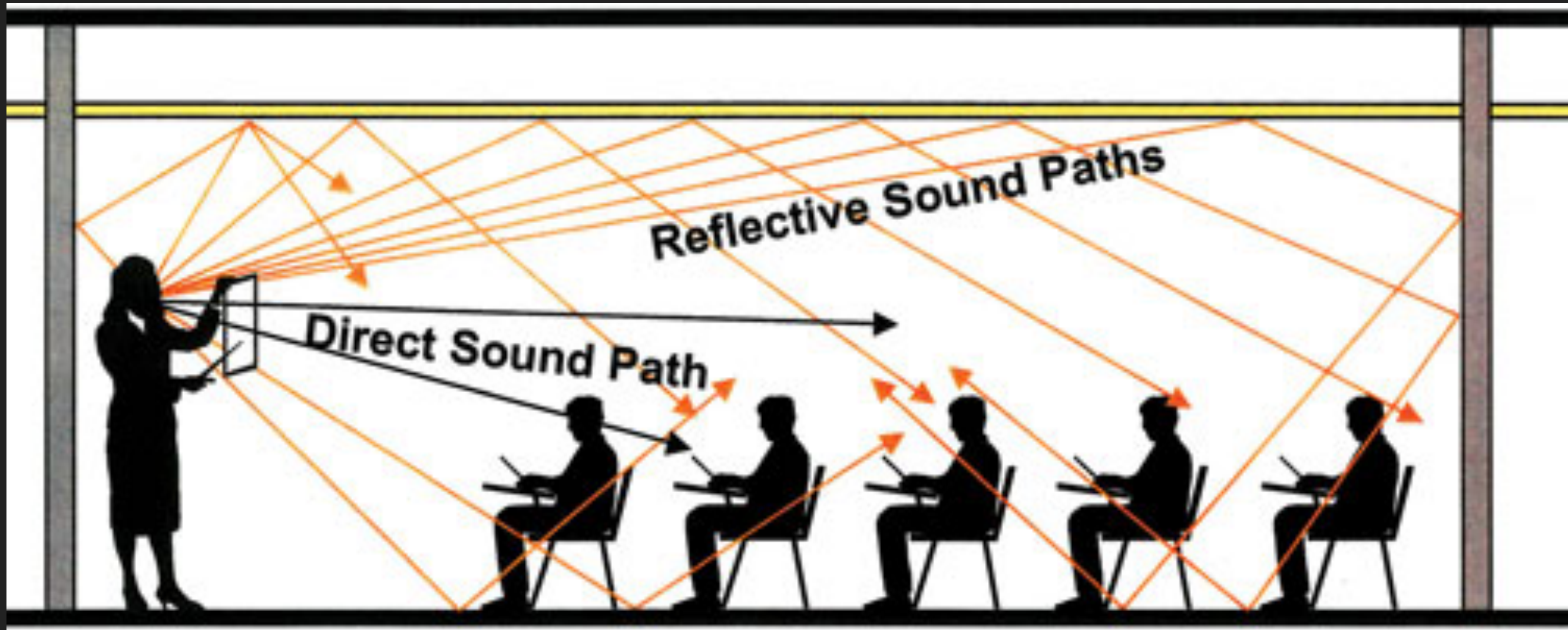


DEFINING SPACE THROUGH SOUND

- ▶ 1. Space- mono, stereo, surround
- ▶ 2. Dynamic range- spl measured in db
- ▶ 3. Frequency/aka pitch
- ▶ 4. Time- how sound evolves in time

DIRECT AND REFLECTED SOUND



DIRECT SOUND

- ▶ Direct sound issues from the source itself
- ▶ When a person is close to us, we hear direct sound including low frequency chest tones.
- ▶ As the person moves farther away, we hear more of the reflected sound.

REFLECTED SOUND

- ▶ Reflected sound is produced by the direct sound bouncing off the walls, floor etc.

REFLECTED SOUND

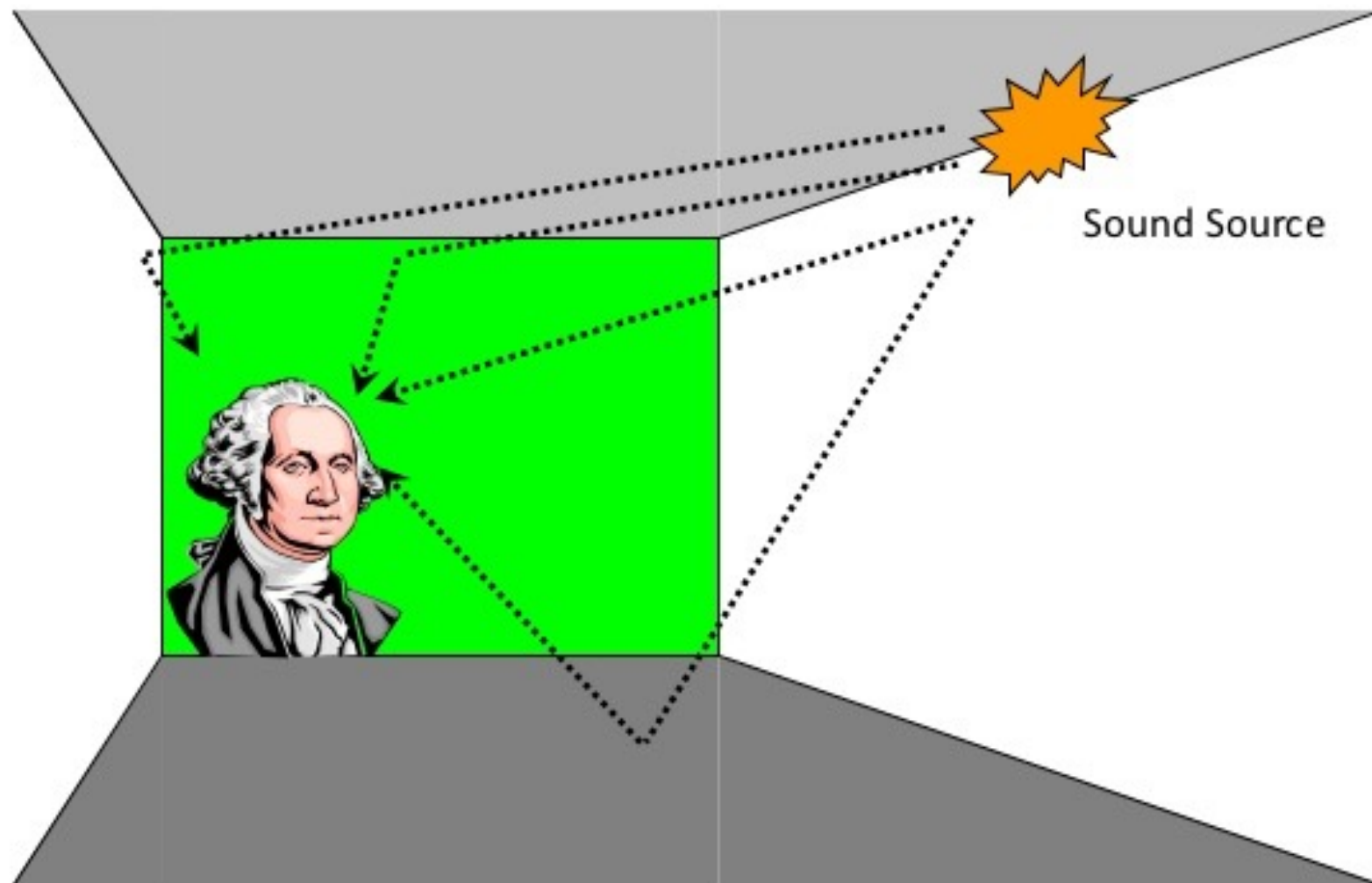
- ▶ **Reflected sound** is more complex in character than direct sound because the surfaces are at different distances from the source and have widely varying reflective properties.

REFLECTED SOUND

- ▶ Hard surfaces - glass, stone, metal, etc. - are said to be "live" because their high reflectivity.
- ▶ Soft or porous materials, like carpeting, draperies and upholstered furniture, sound "dead".

REVERB

- ▶ Reverb = the many reflections of a sound that occur in a room.



Phenomenon of Reverberation

REVERB

- ▶ From any sound source, there is a direct path that the sounds cover to reach our ears. But...sound waves can also take a slightly longer path by reflecting off a wall or the ceiling, before arriving at your ears.

REVERB

- ▶ A reflected sound wave like this will arrive a little later than the direct sound, since it travels a longer distance, and is generally a little weaker, as the walls and other surfaces in the room will absorb some of the sound energy.
- ▶ Reverb creates the 'spaciousness' of a room.

SOUND PERSPECTIVE

- ▶ Close perspective sound contains a high ratio of direct sound to reflected sound.
- ▶ Distant perspective sound contains a high ratio of reflected sound to direct sound.

LET THE RIGHT ONE IN



ROOM TONE

- ▶ A location's "aural fingerprint" - nonspecific sounds on the upper end (usually between 2k and 8k)
- ▶ Each room has a distinct presence of subtle sounds created by the movement of air particles in a particular volume.
- ▶ A microphone placed in two different empty rooms will produce different room tone for each.

ROOM TONE

- ▶ Room tone is often recorded on set/location
- ▶ Room tone is used to match the production sound track so that it may be cut with the track and provide a continuous-sounding background.
- ▶ Room tone smooths out edits and gives a feeling of life in mix. The soundtrack "going dead" would be perceived by the audience not as silence but as a failure of the sound system.

SUBJECTIVE AMBIENCE

- ▶ In The Bourne Ultimatum- in addition to applying reverb to a voice as the convention says for memory - sound designers Per Hallberg and Karen Baker use subjective ambience.
- ▶ They add an intense bass heavy room tone.
- ▶ They color the ambient sound with intense wooshes.
- ▶ They also change the spatial placement of the ambient sound - from L, R, and C speakers to surround speakers.

THE BOURNE ULTIMATUM



EQUALIZATION

- ▶ (EQ) is the process of boosting or cutting certain frequency components in a signal.

ESTABLISHING SOUND

- ▶ Sound that establishes, from the very beginning of a scene, the general character of the surroundings.
- ▶ Establishing sound is usually removed or regularly reduced in volume during dialogue, but may return in the form of reestablishing sound.

THX-1138



HYPER-REAL SOUND

- ▶ Many times, media (games, film) sound is exaggerated compared to sound in real life.
- ▶ Sound that must be heard needs to be over-emphatically stated, just to read through the clutter.
- ▶ Heard in isolation, the recordings seem silly, but heard in context; they assume a more natural balance.
- ▶ The elements that often best illustrate this effect are recorded while watching a picture/video. They are exaggerated from how they would be in reality, both in loudness and in intimacy/detail.